Bellman: putting the tie

Metric spaces, correspondences and the Maximum Theorem

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1 Metric Spaces

Definition 1.1. *A metric space is a set X and a function d called a metric, $d : X \times X \to R$. The metric $d(x, y)$ satisfies the following four properties:*

M1 *Positivity: $d(x, y) \geq 0$ for all $x, y \in X$.*

M2 *Strict positivity: $d(x, y) = 0$ if and only if $x = y$.*

M3 *Symmetry: $d(x, y) = d(y, x)$ for all $x, y \in X$.*

M4 *Triangle inequality: $d(x, y) \leq d(x, z) + d(z, y)$ for all x, y, and $z \in X$.*

Example 1.2. *Consider the space $C_b(X)$, of bounded continuous functions mapping X in the real line $\mathbb{R}$ $f : X \to \mathbb{R}$. Show that the supnorm i.e.*

$$d_\infty(x, y) = \sup_{t\in X} |x(t) - y(t)| \tag{1}$$

is a metric.

Definition 1.3. *A sequence $\{x_n\}$ in a metric space (X, d) is said to be a Cauchy sequence if for each $\epsilon > 0$ there exists an $N(\epsilon)$ such that $d(x_n, x_m) < \epsilon$ for any $n, m \geq N(\epsilon)$. Thus a sequence $\{x_n\}$ is said to be Cauchy if* $\lim_{n,m\to\infty} d(x_n, x_n) = 0$.

We also have the following definition of convergence:

Definition 1.4. *A sequence $\{x_n\}$ in a metric space (X, d) is said to converge to a limit $x_0 \in X$ if for every $\epsilon > 0$ there exists an $N(\epsilon)$ such that $d(x_n, x_0) < \epsilon$ for $n \geq N(\epsilon)$.*

The following lemma asserts that every convergent sequence in (X, d) is a Cauchy sequence:

Lemma 1.5. *Let $\{x_n\}$ be a convergent sequence in a metric space (X, d). Then $\{x_n\}$ is a Cauchy sequence.*

Proof. Fix any $\epsilon > 0$. Let $x_0 \in X$ be the limit of $\{x_n\}$. Then for all m, n one has

$$d(x_n, x_m) \leq d(x_n, x_0) + d(x_m, x_0)$$

by virtue of the triangle inequality. Because x_0 is the limit of $\{x_n\}$, there exists an N such that $d(x_n, x_0) < \epsilon/2$ for $n \geq N$. Together with the preceding inequality, this statement implies that $d(x_n, x_m) < \epsilon$ for $n, m \geq N$. Therefore, $\{x_n\}$ is a Cauchy sequence. □

Definition 1.6. *A metric space (X, d) is said to be complete if each Cauchy sequence in (X, d) is a convergent sequence in (X, d). That is, in a complete metric space, each Cauchy sequence converges to a point belonging to the metric space.*

Theorem 1.7. *Let $(B(X), \|\cdot\|_\infty)$ be the space of bounded real-valued functions with the sup norm. This space is complete.*

Proof. We claim that if f_n is a Cauchy sequence in $\|\cdot\|_\infty$ then its pointwise limit is its limit and in $B(X)$, i.e. it's a real-valued bounded function: Since for fixed x, $f_n(x)$ is a Cauchy sequence in $\mathbb{R}$ and since $\mathbb{R}$ is complete its limit is in $\mathbb{R}$ and hence the pointwise limit $f(x) = \lim_{n\to\infty} f_n(x)$ is a real-valued function. It is also bounded: Let N be such that for $n, m \geq N$ we have $\|f_n - f_m\|_\infty < \frac{1}{2}$. Then for all x

$$|f(x)| \leq |f(x) - f_N(x)| + |f_N(x)| \leq \|f - f_N\|_\infty + \|f_N\|_\infty$$

where $\|f - f_N\|_\infty \leq \frac{1}{2}$ since for $n \geq N$, $|f_n(x) - f_N(x)| < \frac{1}{2}$ for all x and hence $|f(x) - f_N(x)| = |\lim_{n\to\infty} f_n(x) - f_N(x)| = \lim_{n\to\infty} |f_n(x) - f_N(x)| \leq \frac{1}{2}$ (not <!) for all x and hence $\sup_x |f(x) - f_N(x)| = \|f - f_N\|_\infty \leq \frac{1}{2}$. To finish the proof we need to show f_n converges in norm, i.e. $\|f_N - f\|_\infty \overset{N\to\infty}{\longrightarrow} 0$: Let $\varepsilon > 0$. Let N be such that for $n, m \geq N$ we have $\|f_n - f_m\|_\infty < \varepsilon$. Then for all $n \geq N$

$$|f(x) - f_n(x)| = \lim_{m\to\infty} |f_m(x) - f_n(x)| \leq \varepsilon$$

for all x and hence $\|f - f_n\|_\infty \leq \varepsilon$. □

Definition 1.8. *A function f mapping a metric space (X, d) into itself is called an operator.*

We need a notion of continuity of an operator.

Definition 1.9. *Let $f : X \to X$ be an operator on a metric space (X, d). The operator f is said to be continuous at a point $x_0 \in X$ if for every $\epsilon > 0$ there exists a $\delta > 0$ such that $d[f(x), f(x_0)] < \epsilon$ whenever $d(x, x_0) < \delta$. The operator f is said to be continuous if it is continuous at each point $x \in X$.*

Definition 1.10. *Let (X, d) be a metric space and let $f : X \to X$. We say that f is a contraction or contraction mapping if there is a real number $k, 0 \leq k < 1$, such that*

$$d[f(x), f(y)] \leq kd(x, y) \quad \text{for all} \quad x, y \in X.$$

It follows directly from the definition that a contraction mapping is a continuous operator.

Theorem 1.11 (Contraction Mapping)**.** *Let (X, d) be a complete metric space and let $f : X \to X$ be a contraction. Then there is a unique point $x_0 \in X$ such that $f(x_0) = x_0$. Furthermore, if x is any point in X and $\{x_n\}$ is defined inductively according to $x_1 = f(x), x_2 = f(x_1), \ldots, x_{n+1} = f(x_n)$, then $\{x_n\}$ converges to x_0.*

Proof. Let x be any point in X. Define $x_1 = f(x), x_2 = f(x_1), \ldots$ Express this as $x_n = f^n(x)$. To show that the sequence x_n is Cauchy, first assume that $n > m$. Then

$$\begin{aligned} d(x_n, x_m) &= d[f^n(x), f^m(x)] = \\ & d[f^m(x_{n-m}), f^m(x)] \\ & \leq kd\left[f^{m-1}(x_{n-m}), f^{m-1}(x)\right] \end{aligned}$$

By induction, we get

$$(*) \quad d(x_n, x_m) \leq k^m d(x_{n-m}, x).$$

When we repeatedly use the triangle inequality, the preceding inequality implies that

$$\begin{aligned} & d(x_n, x_m) \leq \\ & k^m [d(x_{n-m}, x_{n-m-1}) + \ldots + d(x_1, x)]. \end{aligned}$$

Applying (*) gives

$$d(x_n, x_m) \leq k^m \left(k^{n-m-1} + \ldots + k + 1\right) d(x_1, x)$$

Because $0 \leq k < 1$, we have

$$d(x_n, x_m) \leq \frac{k^m}{1-k} d(x_1, x) \quad (\dagger)$$

The right side of (†) can be made arbitrarily small by choosing m sufficiently large. Therefore, $d(x_n, x_m) \to 0$ as $n, m \to \infty$. Thus $\{x_n\}$ is a Cauchy sequence. Because (X, d) is complete, $\{x_n\}$ converges to an element of (X, d). The limit point x_0 of $\{x_n\} = \{f^n(x)\}$ is a fixed point of f. Because f is continuous, $\lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n)$. Now $f(\lim_{n\to\infty} x_n) = f(x_0)$ and $\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} x_{n+1} = x_0$. Therefore $x_0 = f(x_0)$.

To show that the fixed point x_0 is unique, assume the contrary. Assume that x_0 and $y_0, x_0 \neq y_0$, are two fixed points of f. But then

$$\begin{aligned} 0 < d(x_0, y_0) = d[f(x_0), f(y_0)] \leq \\ kd(x_0, y_0) < d(x_0, y_0) \end{aligned}$$

which is a contradiction. Therefore f has a unique fixed point. □

Definition 1.12. *Let X be a space of functions, and let $x, y \in X$. Then $x \geq y$ if and only if $x(t) \geq y(t)$ for every t in the domain of the functions.*

Theorem 1.13 (Blackwell's Sufficient Conditions for T to be a Contraction)**.** *Let T be an operator on a metric space (X, d_∞), where X is a space of functions. Assume that T has the following two properties:*

1. *Monotonicity: For any $x, y \in X, x \geq y$ implies $T(x) \geq T(y)$.*
2. *Discounting: Let c denote a function that is constant at the real value c for all points in the domain of definition of the functions in X. For any positive real c and every $x \in X, T(x + c) \leq T(x) + \beta c$ for some β satisfying $0 \leq \beta < 1$. Then T is a contraction mapping with modulus β.*

Proof. For all $x, y \in X, x \leq y + d(x, y)$. Applying properties (a) and (b) to this inequality gives

$$T(x) \leq T(y + d(x, y)) \leq T(y) + \beta d(x, y).$$

Exchanging the roles of x and y and using the same logic implies

$$T(y) \leq T(x) + \beta d(x, y).$$

Combining these two inequalities gives $|T(x) - T(y)| \leq \beta d(x, y)$ or

$$d(T(x), T(y)) \leq \beta d(x, y).$$

$\square$

2 Compactness

As it its well known by the **Heine Borel** theorem in $\mathbb{R}$ each closed and bounded subset $[a, b]$ *i.e.* a closed and bounded subset. We should now explore this definition and the previously exposed theorem. To do so we need two *topological notions*: we define an *open cover* of a set A the collection of open sets $\{G_i\}_{i \in I}$ such that $A \subseteq \bigcup_{i \in I} G_i$. In turn, a finite subcover of the open cover $\{G_i\}_{i \in I}$ is a finite collection of sets, say $\{G_1, \ldots, G_n\}$, taken from the open cover $\{G_i\}_{i \in I}$ that still are able to cover the set A, that is, such that $A \subseteq \bigcup_{i=1}^{n} G_i$. For pleasure we remember that an *open set* is a set such that $A = int\ A$[1]. So now we need to ask a simple question: how to reconcile the fact that $[a, b] \subseteq \mathbb{R}$ is a compact set with the notion of *finite subcover*? Here the answer:

Theorem 2.1. *Each open cover of a closed and bounded interval* $[a, b]$ *has a finite subcover.*

Before attacking the proof of this theorem is helpful to prove the following lemma:

Lemma 2.2. *Let* $\{[a_n, b_n]\}_{n \geq 1}$ *be a collection of closed and bounded intervals of* $\mathbb{R}$ *with* $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ *for each* $n \geq 1$. *We have* $\bigcap_{n \geq 1} [a_n, b_n] \neq \emptyset$.

Proof. Given the collection $\{[a_n, b_n]\}_{n \geq 1}$, let $A = \{a_1, a_2, \ldots, a_n, \ldots\}$. We have $A \subseteq [a_1, b_1]$ and therefore A is a bounded set. By the completeness of $\mathbb{R}$, we can set $x = \sup A$. Since each b_n is an upper bound for A, we have $b_n \geq x$ for each $n \geq 1$. Hence, $a_n \leq x \leq b_n$ for each $n \geq 1$, and therefore $x \in \bigcap_{n \geq 1} [a_n, b_n]$. This implies $\bigcap_{n \geq 1} [a_n, b_n] \neq \emptyset$, as desired. □

We can now prove 2.2:

Proof. Suppose per contra that there exists an open cover $\{G_i\}_{i \in I}$ of $[a, b]$ that does not contain any finite subcover of $[a, b]$. Let $\delta = b - a$ and $c_1 = (a+b)/2$. The collection $\{G_i\}_{i \in I}$ is an open cover also of the intervals $[a, c_1]$ and $[c_1, b]$. Therefore, at least one of these two intervals has no finite subcover of $\{G_i\}_{i \in I}$. Otherwise, from $[a, b] = [a, c_1] \cup [c_1, b]$ it would follow that $[a, b]$ itself would have such a subcover. Wlog, suppose therefore that $[a, c_1]$ has no finite subcover of $\{G_i\}_{i \in I}$. Set $c_2 = (a + c_1)/2$. By repeating the argument just seen, we can assume that also $[a, c_2]$ does not have not such a finite subcover. By proceeding in this way we can construct a collection of intervals $\{[a, c_n]\}_{n \geq 1}$ such that $[a, c_{n+1}] \subseteq [a, c_n]$ and $c_n - a = \delta/2^n$ for each $n \geq 1$. Moreover, none of these closed intervals has a finite subcover of $\{G_i\}_{i \in I}$.

By Lemma 2.2, $\bigcap_{n \geq 1} [a, c_n] \neq \emptyset$. Let $x \in \bigcap_{n \geq 1} [a, c_n]$. Since $[a, b] \subseteq \bigcup_{i=1}^{n} G_i$, there exists G_i such that $x \in G_i$. As x is an interior point of G_i, there exists a neighborhood $(x - \varepsilon, x + \varepsilon)$ such that $(x - \varepsilon, x + \varepsilon) \subseteq G_i$. For n sufficiently large, we have $\delta/2^n < \varepsilon$, and therefore

$$[a, c_n] \subseteq \left(x - \frac{\delta}{2^n}, x + \frac{\delta}{2^n} \right) \subseteq (x - \varepsilon, x + \varepsilon) \subseteq G_i.$$

Consequently, the singleton $\{G_i\}$ is a finite subcover of $\{G_i\}_{i \in I}$ that covers $[a, c_n]$, which contradicts the fact that all the intervals $[a, c_n]$ do not have such subcovers. From this contradiction it follows that $[a, b]$ has a finite subcover of $\{G_i\}_{i \in I}$. □

Definition 2.3. *A subset A of a metric space is compact if each open cover of A has a finite subcover.*

Theorem 2.4. *Compact subsets of a metric space are closed and bounded*

Proof. Let K be a compact set of a metric space X. We prove that K^c is open. Let $x \in K^c$. For each $y \in K$, let G_y and V_y be, respectively, neighborhoods of y and x with radius lower than $d(x, y)/2$. This implies that $G_y \cap V_y = \emptyset$ for each $y \in K$. Since the collection $\{G_y\}_{y \in K}$ is an open cover of K, there exists a finite subcover $\{G_{y_i}\}_{i=1}^{n}$ of K. Setting $V = \bigcap_{i=1}^{n} V_{y_i}$ and $G = \bigcup_{i=1}^{n} G_{y_i}$, we have $K \subseteq G$ and $V \cap G = \emptyset$.

[1]Given a set $A \subseteq X$ a point $x \in X$ is called an interior point of A if there exists a neighborhood $B_\varepsilon(x)$ included in A, that is, $B_\varepsilon(x) \subseteq A$ (the set of all interior points of A, denoted by int A, is called the interior of A);

Hence, V is a neighborhood of x such that $V \subseteq K^c$, which implies that x is an interior point of K^c. Thus, K^c is open.

It remains to show that K is bounded. Given $\varepsilon > 0$, for each $x \in K$ let $B_\varepsilon(x)$ be a neighborhood of radius ε. Since the collection $\{B_\varepsilon(x)\}_{x\in K}$ is an open cover of K, there exists a finite set $E \subseteq K$ such that $\{B_\varepsilon(x)\}_{x\in E}$ is a finite subcover of K, that is, $K \subseteq \bigcup_{x\in E} B_\varepsilon(x_i)$. Set $M = \max_{x',x''\in E} d(x', x'')$. Let $y', y'' \in K$. There exist $x', x'' \in E$ such that $y' \in B_\varepsilon(x')$ and $y'' \in B_\varepsilon(x'')$. Therefore,

$$\begin{aligned} d(y', y'') &\leq d(y', x') + d(x', x'') + d(x'', y'') \\ &< 2\varepsilon + M \end{aligned}$$

It follows that, taking any $y \in K$, we have $K \subseteq B_{2\varepsilon+M}(y)$, which implies that K is a bounded set. □

Theorem 2.5. *(Heine-Borel) Let $X = \mathbb{R}^n$, endowed with any of the metrics d_1, d_2 and d_∞. A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. In view of 2.4, we only have to prove that a closed and bounded subset of $\mathbb{R}^n$ is compact. The special case $X = \mathbb{R}$ is given by 2.2. We leave to the reader the proof of the general case $X = \mathbb{R}^n$. □

Theorem 2.6. *In a metric space the closed subsets of a compact set are themselves compact.*

Proof. Let F be a closed subset of a compact set K. Let $\{G_i\}_{i\in I}$ be an open cover of F. We want to prove that there exists a finite subcover of F. Since F is closed, the complement F^c is open. Therefore, the collection $\{G_i\}_{i\in I} \cup \{F^c\}$ is an open cover of K. Therefore, there exists a finite subcover $\{G_i\}_{i=1}^n \cup \{F^c\}$ of K, i.e., $K \subseteq F^c \cup \bigcup_{i=1}^n G_i$. This implies $F \subseteq \bigcup_{i=1}^n G_i$, and therefore $\{G_i\}_{i=1}^n$ is a finite subcover of F. □

Theorem 2.7. *A subset A of a metric space is compact if and only if it has the Bolzano-Weierstrass property, that is, each sequence of points of A has at least a subsequence that converges to some point of A.*

Bolzano-Weierstrass property is often referred as the sequential compactness property, which is thus equivalent to compactness in metric spaces.

Proof. We only prove the "only if" part. Suppose that A is compact. We want to show that any sequence $\{x_n\}$ of points of A has at least a subsequence that converges to some point x of A.

First, suppose that the range of the sequence is finite, that is, the terms of the sequence take on finitely many values of A. At least one of such values, say $\bar{x}$, must occur infinitely many times within the sequence. Hence, there is a subsequence which is constant to $\bar{x}$, and so trivially converges to $\bar{x} \in A$.

Suppose now that the sequence takes on infinitely many (distinct) values of A. We first show that there exists some point of the set A such that all its neighborhoods contain infinitely many points of the sequence $\{x_n\}$. Suppose, per contra, that for each $y \in A$ there exists a neighborhood $B_{\varepsilon_y}(y)$ that contains at most a finite number of elements of $\{x_n\}$. The family $\left\{B_{\varepsilon_y}(y)\right\}_{y\in A}$ is an open cover of A, and so, being A compact, there exists a finite set $\{y_i\}_{i=1}^n$ such that $A \subseteq \bigcup_{i=1}^n B_{\varepsilon_{y_i}}(y)$. Since each set $B_{\varepsilon_{y_i}}(y)$ contains at most a finite number of elements of the sequence $\{x_n\}$, there are at most a finite number of such elements that belong to $\bigcup_{i=1}^n B_{\varepsilon_{y_i}}(y)$. Since $A \subseteq \bigcup_{i=1}^n B_{\varepsilon_{y_i}}(y)$, this contradicts the fact that $\{x_n\}$ takes on infinitely many values in A. We conclude that there is a point $x \in A$ such that all its neighborhoods $B_\varepsilon(x)$ contain contains infinitely many values of the sequence $\{x_n\}$. Define recursively a sequence $\{n_k\}$ as follows:

$$\begin{aligned} n_1 &= \min\{n \geq 1 : x_n \in B_1(x)\} \\ n_2 &= \min\left\{n \geq n_1 : x_n \in B_{\frac{1}{2}}(x)\right\} \\ &\ \ \cdots \\ n_k &= \min\left\{n \geq n_{k-1} : x_n \in B_{\frac{1}{k}}(x)\right\} \\ &\ \ \cdots \end{aligned}$$

Clearly, the subsequence $\{x_{n_k}\}$ converges to $x \in A$, as desired. □

3 A primer on correspondences

Given any two sets X and Y, a correspondence is a multivalued function $\varphi : D \subseteq X \rightarrow 2^Y$ with $\varphi(x) \neq \emptyset$ for all $x \in D$. In other words, with each element $x \in D$ is associated a nonempty subset $\varphi(x)$ of Y (the image of x under φ). The set D is the domain of φ, written $\operatorname{dom} \varphi$, and Y is the codomain. If $D = X$, we say that φ is viable.

When $\varphi(x)$ is a singleton for all $x \in D$, the correspondence reduces to a function. In particular, all functions $f : D \subseteq X \rightarrow Y$ such that $f(x) \in \varphi(x)$ for all $x \in D$ are called selections of φ. A correspondence is a function if and only if it has a unique selection.

Example 3.1. *The budget correspondence* $B : \mathbb{R}_+^{n+1} \rightarrow 2^{\mathbb{R}_+^n}$ *is given by the budget set*

$$B(p, w) = \left\{x \in \mathbb{R}_+^n : p \cdot x \leq w\right\}$$

Note that $B(p, w) \neq \emptyset$ *for all* $(p, w) \in \mathbb{R}_+^{n+1}$ *since* $\mathbf{0} \in B(p, w)$ *for all* $(p, w) \in \mathbb{R}_+^{n+1}$. □

If Y is a metric space, we say that φ is: (i) closed-valued if all its images $\varphi(x)$ are closed subset; (ii) compact-valued if all its images $\varphi(x)$ are compact subset. If Y is a vector space, we say that φ is: (iii) convex-valued if all its images $\varphi(x)$ are convex subsets.

Definition 3.2. *The graph* $Gr\ \varphi$ *of a correspondence* $\varphi : D \subseteq X \rightarrow 2^Y$ *is the set*

$$\operatorname{Gr} \varphi = \{(x, y) \in D \times Y : y \in \varphi(x)\} \subseteq X \times Y$$

Definition 3.3. *Suppose that* X *and* Y *are metric spaces. There are several notions of continuity for correspondences and each can be characterized in a variety of ways. The following definitions are useful in metric spaces. A correspondence* $\varphi : D \subseteq X \rightarrow 2^Y$ *is*

1. *upper hemicontinuous at* $x \in D$ *if*

$$x_n \rightarrow x \text{ and } y_n \in \varphi\left(x_n\right)$$

implies the existence of a subsequence $\left\{y_{\pi_k}\right\}$ *that converges to some* $y \in \varphi(x)$.

2. *lower hemicontinuous at* $x \in D$ *if*

$$x_n \rightarrow x \text{ and } y \in \varphi(x)$$

implies that there exist a subsequence $\left\{x_{n_k}\right\}$ *and elements* $y_k \in \varphi\left(x_{n_k}\right)$ *such that* $y_k \rightarrow y$.

3. *continuous at* $x \in D$ *if it is both upper and lower hemicontinuous at* x.

A correspondence φ *is upper (lower) hemicontinuous if is upper (lower) hemicontinuous at all* $x \in \operatorname{dom} \varphi$.

Definition 3.4. *A correspondence* $\varphi : X \rightarrow 2^Y$ *is closed at* x *if* $x_n \rightarrow x$ *and* $y_n \rightarrow y$, *with* $y_n \in \varphi\left(x_n\right)$, *implies* $y \in \varphi(x)$.

Theorem 3.5. *If* $\varphi : D \subseteq X \rightarrow 2^Y$ *is upper hemicontinuous at* $x \in D$, *then* φ *is closed at* x *and its image* $\varphi(x)$ *is compact.*

Theorem 3.6. *For a function* $f : D \subseteq X \rightarrow Y$ *and a point* $x \in D$, *the following properties are equivalent:*

1. f *is continuous at* x;
2. f *is lower hemicontinuous at* x;
3. f *is upper hemicontinuous at* x.

Proof. First observe that, being f a function, $y \in f(x)$ amounts to $y = f(x)$. In view of this observation, it is easy to check that (i) implies (ii) and (ii) implies (iii). [2]

Let us show that (iii) implies (i). Let $x_n \rightarrow x$. We want to show that $y_n = f\left(x_n\right) \rightarrow y = f(x)$. Suppose not. Then there is $\varepsilon > 0$ and a subsequence $\left\{y_{n_k}\right\}$ such that

$$d\left(y_{n_k}, y\right) \geq \varepsilon \quad \forall k$$

Since $x_{n_k} \rightarrow x$, by upper hemicontinuity there is a further subsequence $\left\{y_{n_{k_s}}\right\}$ that converges to $y = f(x)$. Hence, for all s large enough $d\left(y_{n_{k_A}}, y\right) < \varepsilon$, which contradicts (12.1). We conclude that $y_n = f\left(x_n\right) \rightarrow y = f(x)$, i.e., f is continuous at x.

□

4 Maximum Theorem

Consider a set of parameters Θ and a set X of choice variables, suppose that each value θ of the parameter determines a feasible set $\varphi(\theta) \subseteq X$. That is, feasible sets are identified by a *feasibility correspondence* $\varphi : \theta \to 2^X$. Consider an *objective function* f defined over the graph of the correspondence φ; *i.e.* $f : Gr\phi \to \mathbb{R}$. This objective function has to be optimized on the feasible sets determined by the correspondence $\varphi : \Theta \to 2^X$. Jointly, φ and f determines an optimization problem in parametric form:

$$\max_x f(x, \theta) \quad \text{sub } x \in \varphi(\theta) \tag{2}$$

When f is concave in x and φ is convex-valued, problem (2) is called concave. If problem (2) admits solutions for $\theta \in \Theta$, there is $\hat{x} \in \varphi(\theta)$ such that $f(\hat{x}, \theta) \geq f(x, \theta)$ for each $x \in \varphi(\theta)$. In this way, key functions and correspondences may be introduced. In particular, the solution correspondence $\sigma : D \subseteq \Theta \to 2^X$ of problem (2) is given by

$$\sigma(\theta) = \arg \max_{x \in \varphi(\theta)} f(x, \theta)$$

That is, the correspondence σ collects all solutions of problem (2). Its domain, $\operatorname{dom} \sigma = D$, is the solution domain, that is, the collection of all θ for which problem (2) admits a solution. Such solution is unique at all $\theta \in \operatorname{dom} \sigma$ when σ is single-valued, that is, when σ is a function. In this case we say that σ is the solution function of problem (2). A last key function associated with problem (12.2) is the (optimal) value function $v : \operatorname{dom} \sigma \subseteq \Theta \to \mathbb{R}$ given by

$$v(\theta) = \max\{f(x, \theta) : x \in \varphi(\theta)\} = f(\hat{x}, \theta) \qquad \forall \hat{x} \in \sigma(\theta)$$

for each $\theta \in \operatorname{dom} \sigma$. The value function gives the value of the objective function when optimized. As such, it is well defined only on the solution domain $\operatorname{dom} \sigma$.

The next important result is known as Berge's Maximum Theorem.

Theorem 4.1. *Consider a parametric optimization problem (2) in which X and Θ are metric spaces.*

If φ is upper hemicontinuous and f is upper semicontinuous, then $\operatorname{dom} \sigma = \Theta$ *and v is upper semicontinuous.*

If, in addition, φ is lower hemicontinuous and f is lower semicontinuous (and so both are continuous), then v is continuous and σ is upper hemicontinuous.

The proof relies on couple of lemmas of independent interest. The first reports a property of sequences.

Lemma 4.2. *Given any sequence of scalars $\{a_n\}$, if* $\limsup_{n\to+\infty} a_n = a \in [-\infty, +\infty]$, *then there is a subsequence $\{a_{n_k}\}$ such that* $\lim_{k\to+\infty} a_{n_k} = a$.

Proof. If for each $n \geq 1$ we set $A_n = \sup_{m\geq n} a_m \in [-\infty, +\infty]$, it holds $a = \lim_{n\to+\infty} A_n = \inf_n A_{n+}$ Suppose $a \in \mathbb{R}$. Fix any $\varepsilon > 0$. There is some n_ε such that $A_{n_\varepsilon} - a < \varepsilon/2$. By the definition of supremum, there is some $m \geq n_\varepsilon$ such that $A_{n_e} - \varepsilon/2 \leq a_m \leq A_{n_e}$. In turn, this easily implies

$$|a_m - a| =$$

$$|a_m - A_{n_\epsilon} + A_{n_\varepsilon} - a| \leq |a_m - A_{n_\varepsilon}| + |A_{n_\varepsilon} - a| < \varepsilon$$

Given any integer $k \geq 1$, set $\varepsilon = 1/k$ and denote by a_{n_n} the term of the sequence $\{a_n\}$ such that $|a_{n_k} - a| < 1/k$. In this way we construct a subsequence $\{a_{n_k}\}$ that converges to a.

We leave to the reader to check that if $a = \pm\infty$ there is a subsequence $\{a_{n_k}\}$ that diverges to $\pm\infty$, respectively. □

A similar property holds for the liminf. The next lemma shows how upper hemicontinuity can be inherited through inclusion.

Lemma 4.3. *Let $\varphi, \psi : X \to 2^Y$ be two correspondences such that φ is closed and $\varphi(x) \subseteq \psi(x)$ for each $x \in X$. If ψ is upper hemicontinuous, then φ is also upper hemicontinuous.*

Proof. Let $x_n \to x$ and $y_n \in \varphi(x_n)$. As $\varphi(x_n) \subseteq \psi(x_n)$ for each n and ψ is upper hemicontinuous, there exists a subsequence $\{y_{n_k}\}$

such that $y_{n_k} \to y \in \psi(x)$. As φ is closed, from $x_{n_k} \to x, y_{nk} \to y$ and $y_{n_k} \in \varphi\left(x_{n_k}\right)$, it follows $y \in \varphi(x)$, as desired. $\square$

Proof of the Maximum Theorem. Suppose φ is upper hemicontinuous and f is upper semicontinuous. By Theorem 3.5, $\varphi(\theta)$ is compact for each $\theta \in \Theta$. Since f is upper semicontinuous, by the Weierstrass Theorem the problem (12.2) has a solution, i.e., there is $\hat{x} \in \varphi(\theta)$ such that $f(\hat{x}, \theta) \geq f(x, \theta)$ for all $x \in \varphi(\theta)$. Hence, $\operatorname{dom} \sigma = \Theta$.

Let us prove that $v : \theta \to \mathbb{R}$ is upper semicontinuous. Fix any point $\bar{\theta} \in \Theta$ and consider a sequence $\{\theta_n\} \subseteq \Theta$ such that $\lim_{n\to\infty} \theta_n = \bar{\theta}$. We want to show that $\lim_{\sup_{n\to\infty}} v\left(\theta_n\right) \leq v(\bar{\theta})$. Set $\limsup_{n\to\infty} v\left(\theta_n\right) = \lambda \in [-\infty, +\infty]$. By Lemma 4.2, there exists a subsequence $\{\theta_{nk}\}$ such that $\lim_{k\to\infty} v\left(\theta_{n_k}\right) = \lambda$. Let $\hat{x}_k \in \varphi\left(\theta_{n_k}\right)$ for each $k \geq 1$, so that $f\left(\hat{x}_k, \theta_{n_k}\right) = v\left(\theta_{n_k}\right)$ for each $k \geq 1$. Since φ is upper hemicontinuous and $\lim_{k\to\infty} \theta_{n_k} = \bar{\theta}$, there is a subsequence $\{\hat{x}_{k_n}\}$ that converges to some $\bar{x} \in \varphi(\bar{\theta})$. Since f is upper hemicontinuous, this implies

$$\lambda = \lim_{\bar{s}\to\infty} v\left(\theta_{n_k}\right) = \limsup_{s\to\infty} v\left(\theta_{n_{k_s}}\right) = \limsup_{s\to\infty} f\left(x_{k_s}, \theta_{n_{k_s}}\right) \leq f(\bar{x}, \bar{\theta}) = v(\bar{\theta})$$

We conclude that $\lambda \leq v(\bar{\theta})$, as desired. Suppose now that both f and φ are continuous. Fix any point $\bar{\theta} \in \Theta$ and consider a sequence $\{\theta_n\} \subseteq \Theta$ such that $\lim_{n\to\infty} \theta_n = \bar{\theta}$. We first show that v is continuous at $\bar{\theta}$. Since v is upper semicontinuous, it is enough to prove that it is lower semicontinuous, i.e., $\liminf_{n\to\infty} v\left(\theta_n\right) \geq v(\bar{\theta})$. Set $\liminf_{n\to+\infty} v\left(\theta_n\right) = \lambda \in [-\infty, +\infty]$. By Lemma 4.2, there exists a subsequence $\{\theta_{n_k}\}$ such that $\lim_{k\to\infty} v\left(\theta_{n_k}\right) = \lambda$. Since $\operatorname{dom} \sigma = \Theta$, there is $\bar{x} \in \varphi(\bar{\theta})$ such that $v(\bar{\theta}) = f(\bar{x}, \bar{\theta})$. Since φ is lower hemicontinuous, there exist a further subsequence $\theta_{n_{k_a}}$ and elements $x_{k_z} \in \varphi\left(\theta_{n_{k_z}}\right)$ such that $\lim_{s\to\infty} x_{k_n} = \bar{x}$. Since $v\left(\theta_{n_{k_n}}\right) = f\left(x_{k_n}, \theta_{n_{k_s}}\right)$ for each $s \geq 1$, the continuity of f implies

$$\lambda = \lim_{s\to\infty} v\left(\theta_{n_{k_A}}\right) = \lim_{s\to\infty} f\left(x_{k_s}, \theta_{n_{k_s}}\right) = f(\bar{x}, \bar{\theta}) = v(\bar{\theta}).$$

Hence, $\liminf_{n\to\infty} v\left(\theta_n\right) = v(\bar{\theta})$, as desired. $\square$

5 The Bellman equation

$$V(x) = \max_y F(x, y) + \beta V(y)$$
$$\text{s.t.}$$
$$y \in \Gamma(x)$$

- Some terminology:
- The Functional Equation (1) is called a Bellman equation.
$-x$ is called a state variable.
- $G(x) = \{y \in \Gamma(x) : V(x) = F(x, y) + \beta V(y)\}$ is called a policy correspondence. It spells out all the values of y that attain the maximum in the RHS of (1).
- If $G(x)$ is single-valued (i.e. there is a unique optimum), G is called a policy function.
- Questions:

1. Does (1) have a solution?
2. Is it unique?
3. How do we find it?

Define the operator T by

$$T(f)(x) = \sup_y F(x, y) + \beta f(y)$$
$$\text{s.t. } y \in \Gamma(x)$$

V can be defined as a fixed point of T, i.e. a function V such that $T(V)(x) = V(x) \quad \forall x$
- Does T have a fixed point? How do we find it? Note that by Berge's maximum value theorem if $X \subseteq \mathbb{R}^n$ is convex. $\Gamma : X \rightrightarrows X$ is nonempty, compactvalued and continuous and $F : X \times X \to \mathbb{R}$ is continuous and bounded then V is continuous, we can then work on the space of continuous bounded functions with the supmetric. The operator T is defined over this space and it's easy to check that it satisfy the Blackwell's sufficient conditions for it to be a contraction. Solutions to (1) are so given by $V = T(V)$, but since T is a contraction we are sure this point (a function) exist and it's unique. Proof of the contraction mapping theorem also gives us a simple way to find a solution. Pick a generic function in the space of continuous bounded functions and iterate the operator till convergence.